Battery

A battery is a device that consists of two or more galvanic cells connected in series or parallel or both, which converts chemical energy into electrical energy through redox reactions.

Example: Lead-Acid battery, Nickel-Cadmium battery etc.

Basic Components of a Battery

A battery consists of four major components.

1. **Anode (or negative electrode):** It releases electrons into the external circuit by undergoing oxidation. $M \longrightarrow M^{n+} + ne^-$

2. **Cathode (or positive electrode):** It accepts electrons coming from anode through external circuit. Thereby reduction of the active species occurs at cathode.

$$M^{n+} + ne^- \longrightarrow M$$

3. Electrolyte: It provides the medium for transfer of ions inside the cell between the anode and cathode. A solution of an acid, alkali or salt having high ionic conductivity is commonly used as an electrolyte.

4. Separator: It is used to separate anode and cathode compartments in a battery to prevent internal short circuiting. A separator allows transport of ions from anode to cathode and vice versa to maintain the electrical neutrality. It is an electrolytic conductor but an electronic insulator. Example: Cellulose, cellophane, nafion membranes etc.

In addition, cathode current collector, anode current collector, terminals, seal and container make a battery complete.

Classification of Batteries

Batteries are classified as 1) Primary batteries 2) Secondary batteries and 3) Reserve batteries

1. Primary Batteries: A battery which cannot be recharged (Because the cell reactions are irreversible) and discarded when the battery has delivered all its electrical energy is called as a primary battery. **Example:** Zn-MnO_2 battery, Li- MnO_2 battery etc.

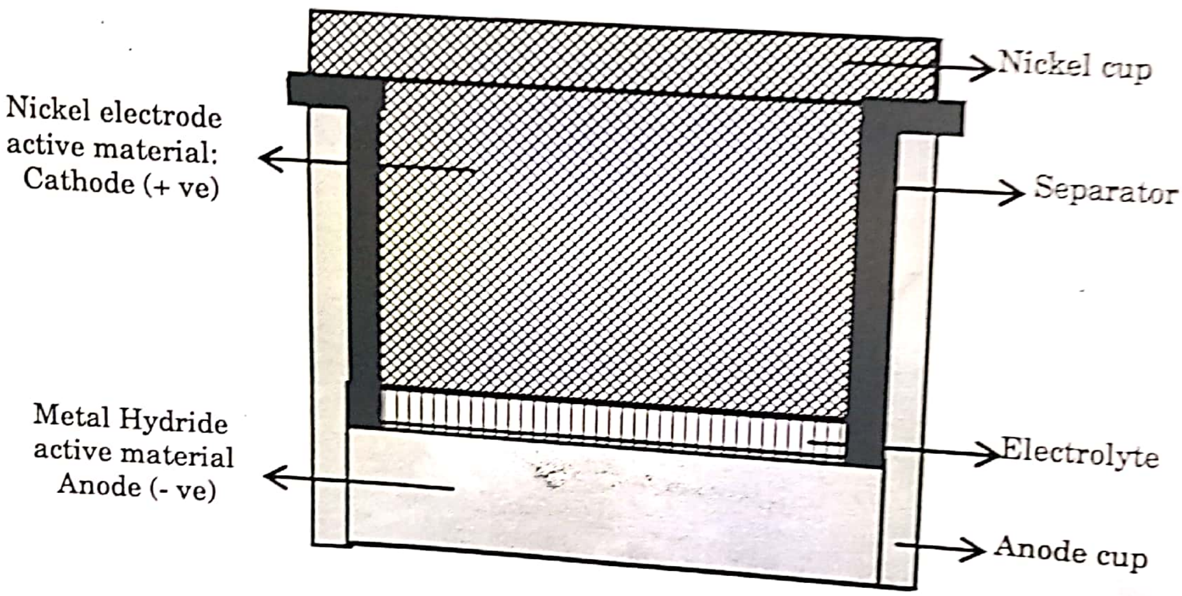
2. Secondary Batteries: A battery which after discharging, can be recharged (As the cell reactions are reversible) again by passing the electric current through it in the opposite direction to that of the discharge is known as the secondary battery. The secondary batteries are also called as storage batteries as they are the storage devices for electrical energy. **Example:** Lead storage battery, Nickel-cadmium battery etc.

3. Reserve Batteries: In reserve batteries, one of the components is stored separately and is incorporated into the battery when required. Usually, the electrolyte is separately stored. **Example:** Mg-AgCl and Mg-CuCl batteries; both can be activated whenever required just by adding water. Reserve batteries are used to deliver high power for relatively short period of time in missiles, torpedoes and other weapon systems.

Construction of Ni – MH Battery:

1. **The Anode:** Porous nickel grid pasted with hydrides of metals like VH_2, ZrH_2 and TiH_2 with a hydrogen storage metal alloy such as $TiNi_2$ or $LaNi_5$.
2. **The Cathode:** Porous nickel grid pasted with $Ni(OH)_2$.
3. **The Electrolyte:** 28 % KOH solution.

The anode and cathode are taken in a stainless steel container and are separated by polypropylene membrane.



Nickel-Metal hydride cell

Cell Representation: $MH_2, M \mid KOH\ (28\%) \mid NiO(OH), Ni(OH)_2$

Reactions:

MH (-1) 0.5 (oxdn)

M (+1) 0.5

Anode : $MH_2 + 2OH^- \underset{\text{Charge}}{\overset{\text{Discharge}}{\rightleftharpoons}} 2H_2O + M + 2e^-$

$MH + OH^{\ominus} \rightleftharpoons M + H_2O + e^{\ominus}$

Cathode : $2NiO(OH) + 2H_2O + 2e^- \underset{\text{Charge}}{\overset{\text{Discharge}}{\rightleftharpoons}} 2Ni(OH)_2 + 2OH^-$

$NiO(OH) + H_2O + e^{\ominus} \rightleftharpoons Ni(OH)_2 + OH^{\ominus}$

Overall reaction : $MH_2 + 2NiO(OH) \underset{\text{Charge}}{\overset{\text{Discharge}}{\rightleftharpoons}} M + 2Ni(OH)_2$

$MH + NiO(OH) \rightleftharpoons M + Ni(OH)_2$

Applications: It is used in cellular phones, laptop computers, electric vehicles and spacecrafts.

Ni-Cd and Ni-MH batteries are manufactured in discharged state, i.e. by taking active materials ($Ni(OH)_2$ in this case) obtained at the end of discharge. Therefore, these batteries should be charged to full capacity before first use.

Lithium-Ion Battery

Rechargeable, Lithium ion batteries possess many advantages over traditional rechargeable batteries, such as lead acid and Ni–Cd batteries. They include high voltage, high energy density (240 W hr/Kg), high cyclic life and very low self-discharge when not in use. It provides maximum voltage of 3.7 V after complete charge.

Construction of Lithium-Ion Battery:

Lithium-ion batteries are composed of three parts: anode, cathode, and an electrolyte. Anode and cathodes are able to insert lithium ions into their layered structure reversibly.

1. The Anode: Lithium inserted in Graphite carbon.

A Little More...

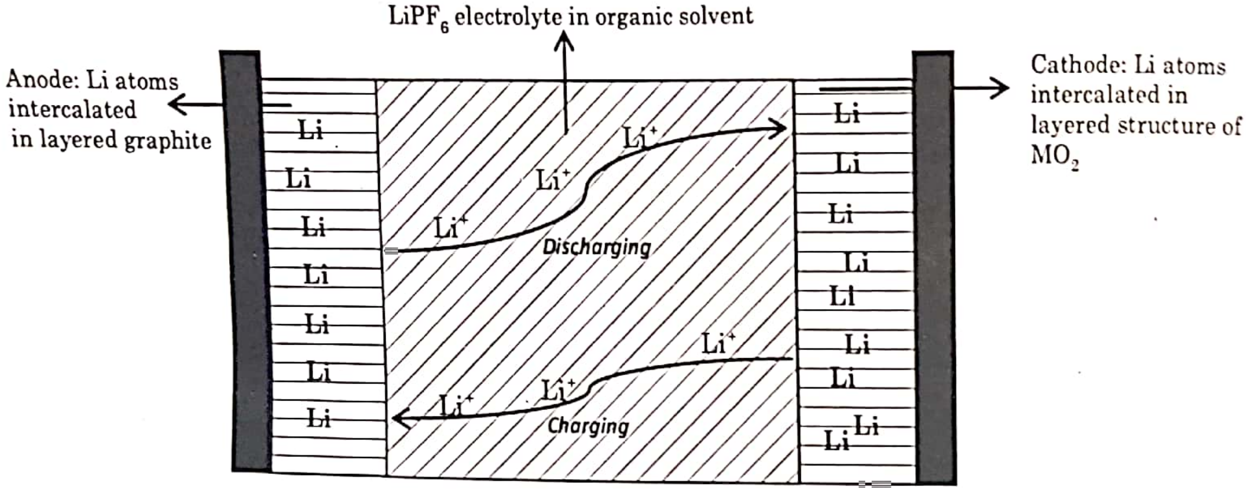
Graphite has a layered structure with different sheets of carbon held together by van der Waals forces. Within each layer, each carbon is bonded to three other carbons through sp^2 orbitals forming hexagons. One lithium can be inserted between two hexagons of consecutive layers. This limits the amount of lithium to one per every six carbon atoms. Due to this, energy density of lithium ion battery is theoretically limited. But, Li atoms that enter into graphite layer during charging come out very easily during discharging. The structure of graphite remains intact during charging and discharging with very low expansion. Thus, it is possible to repeat charging-discharging process many times. Hence, Li-ion batteries have high cycle life.

2. TheCathode: lithium inserted in metal oxide like MnO_2.

3. Electrolyte: A lithium salt such as $LiPF_6$ dissolved in binary organic solvent mixture such as ethylene carbonate-dimethyl carbonate.

Cell Representation: → Li-hexa fluoro phosphate

$$Li \mid Li^+, C \mid LiPF_6 \text{ in ethylene carbonate} \mid Li\text{-}MnO_2$$



Lithium ion rechargeable battery

Reactions: During discharging of battery, Lithium atoms present in graphite layer (one Li atom is present for every 6 C atoms) are oxidized, liberating electrons and lithium ions. Electrons flow through external circuit to cathode and lithium ions flow through the organic electrolyte towards cathode.

$$Li\text{-}C_6 \underset{\text{Charging}}{\overset{\text{Discharging}}{\rightleftharpoons}} Li^+ + 6C + e^-$$

At cathode, lithium ions are reduced to lithium atoms and are inserted in to the layered structure of metal oxide.

$Li^{+} + MnO_2 + e^{-}$ / MO_2 $\underset{\text{Charging}}{\overset{\text{Discharging}}{\rightleftharpoons}}$ $Li\text{-}MnO_2$ / $Li\text{-}MO_2$

$$\boxed{Li\text{-}MnO_2 \underset{\text{Charging}}{\overset{\text{Discharging}}{\rightleftharpoons}} Li^+ + MnO_2 + e^-}$$

Reactions are revesred during charging of battery and Lithium atoms are refilled in to the layered structure of graphite.

Applications: They are currently used for powering portable electronic devices like cell phones, laptops, and also in defence, electrical vehicles, power tools and aerospace applications.

Batteries:-

II) characteristics of batteries:

** i) Voltage (EMF):-

The voltage of a battery is given by the equation

$$\boxed{E_{cell} = (E_C - E_A) - \eta_A - \eta_C - iR_{cell}}$$

Here, E_C & E_A are the electrode potentials of cathode & the anode.

η_A & η_C are the over potential ~~of~~ at anode & cathode.

iR_{cell} is the internal cell resistance.

The emf of the cell depends upon the free energy change in the cell reaction. The emf of the cell is given by Nernst equation,

$$\boxed{E = E^0 - \frac{2.303\,RT}{nF}\log Q}$$

Here, Q = Reaction quotient $= \frac{[\text{Product}]}{[\text{Reactant}]}$

$$E = E^0_{cathode} - E^0_{anode}$$

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2) <u>Capacity</u>:-

Capacity is the charge or the amount of electricity in ampere hours (Ah) that can be obtained from a battery.

→ A good battery must have higher capacity.

→ capacity depends on the size of the battery and it is determined by the Faraday equation.

$$\boxed{C = \frac{wnF}{M}}$$

Here, 'c' is the capacity in Ah.

'w' is the mass of the active material.

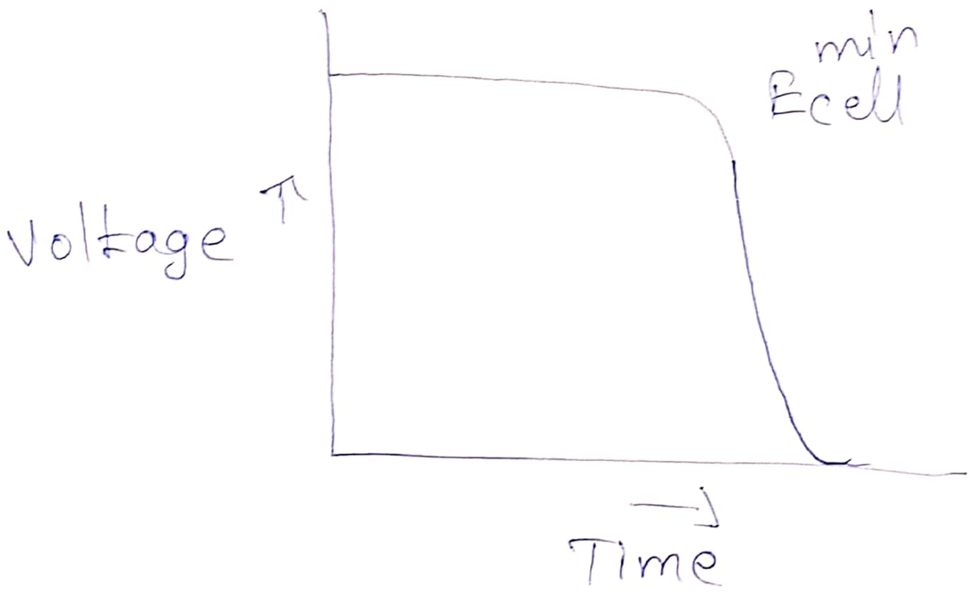
'M' is the molar mass of the active material.

'n' is the number of electrons involved in cell reaction.

→ Capacity of the battery depends upon the amount of active materials consumed during discharge, size of the battery & also on the discharge conditions.

→ capacity of a battery is measured by plotting time against voltage at a fixed current discharge as shown in figure.

→ More is the length of the flat portion of the curve, better is the capacity of the battery.



5. Energy Efficiency: It is the ratio of the output of the battery on the discharge to the input required to restore it to the initial state of charge under specified conditions. It is given by

$$\% \text{ energy efficiency} = \frac{\text{Energy available on discharge}}{\text{Energy required for charging}} \times 100$$

It is pertinent to secondary or storage batteries.

6. Cycle Life: Cycle life of a battery is defined as the number of times the discharging and charging can be repeated before failure of the battery. It is applicable only for the secondary batteries.

7. Shelf Life: It is defined as the duration of storage under specified conditions at the end of which a battery still retains the ability to give a specified performance.

8. Energy Density: Energy density is defined as the ratio of the energy available from a battery to its mass measured in W hr/Kg. It is given by;

$$\text{Energy Density} = \frac{I\,E\,t}{w}$$

Where 'I' is the current drawn from the battery at fixed voltage 'E' for time duration 't' and 'w' is the weight of battery. For major applications, a good battery is expected to have good energy density.

9. Power Density: Power density is the ratio of the power available from a battery to its mass. It is measured in W/Kg. The power density of battery is given by,

$$\text{Power density} = \frac{I\,E}{w}$$

Where 'I' is the current drawn from the battery and E is the voltage and 'w' is weight of battery.

If a battery is operated at high power level, then the energy available will be less. To obtain maximum amount of energy, battery should be operated at a low power level. For few applications like ignition starting of an engine, a battery is expected to have very high power density for a short period.

Q- Why electrolyte used in Lithium batteries should be non-aqueous electrolyte:
Aqueous electrolyte can not be used in Li-batteries because of high reactivity of Li with H_2O.

Q. Explain the construction, reactions and applications of Al- air (Metal-air) battery.

Metal-Air Batteries (Al-Air batteries)
Battery can be quite heavy. This disadvantage prevents batteries from being a source ofenergy in many different appliances and applications where being lightweight is crucial.
An **aluminum air battery** overcomes this issue. It uses air as a cathode, significantly reducing its weight.
In an **aluminum air battery**, aluminum is used as an anode, and air (the oxygen in the air) is used as cathode. This results in the energy density i.e. energy produced per unit weight of the battery – very high compared to other conventional batteries.
Despite this an aluminum air battery is not commercially produced, mainly due to the high production cost of the anode, as well as issues with corrosion of the aluminium anode due to the carbon dioxide in air. Because of this, use of this battery is restricted to mainly military applications.
The major advantages and disadvantages of metal – air batteries are summarized in Table

Advantages	**Disadvantages**
High energy density	Dependent on environmental conditions: • Drying out limits shelf life once opened to air • Electrolyte flooding limits power output
Flat discharge voltage	
Long shelf life (dry storage)	
Nontoxic (on metal use basis)	Limited power density
Low cost (on metal use basis)	Limited operating temperature range

Aluminium–air (Al–air) battery: They have one of the highest energy densities of all batteries, however, their use is not extended due to the existing limitation during the recharge process. So, these batteries are primary cells, i.e., non-rechargeable.
Battery notation:Al/KOH (6M)/Air
Reactions: Aluminium–air batteries (Al–air batteries) produce electricity from the reaction of oxygen in the air with aluminium. The battery consists of Al as anodic material and air as catholic material. The electrolyte used is aqueous KOH solution.